MicroNet Guide & Data Cleaning Pipeline

Files Supported by MicroNet

- Data files: these should be the raw output of user's plate reader.
  - Rows: represent individual wells
  - Columns: timepoint readings
  - Values: OD readings
- You can upload multiple as long as matching # of mapping files.

exampledata

Well	0	600	1200
A1	0.105	0.103	0.102
A2	0.099	0.097	0.097
...	...	...	...
H12	0.073	0.073	0.073

Files Supported by MicroNet

- Mapping files: label individual wells
  - Columns represent assigned variables → allow you to further pool or subset data in later analysis
  - Need to be paired with individual data files
- User can upload multiple as long as matching # of datafiles

Well	Plate_ID	Substrate	Ribotype
A1	2020-06-22-R1	Glucose	RT053
B1	2020-06-22-R1	Glucose	RT053
C1	2020-06-22-R1	Fructose	RT053
D1	2020-06-22-R1	Fructose	RT053
…	…	…	…
E12	2020-06-22-R1	Trehalose	RT078
F12	2020-06-22-R1	Trehalose	RT078
G12	2020-06-22-R1	Water	RT078
H12	2020-06-22-R1		RT078

Files Supported by MicroNet

- Metafile: labels entire plates/datafiles
  - MicroNet will use the Plate_ID column to find the corresponding datafile. It will label all wells in “2020-06-22-R1” with “Species” being “E_randomii”.

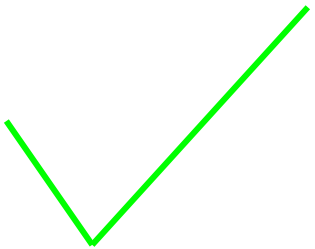
Plate_ID	Species
2020-06-22-R1	E_randomii

- Time interval file: assigns OD reading intervals
  - MicroNet will label “2020-06-22-R1” as having

```
Plate_ID:Interval(seconds)
2020-06-22-R1:1200
2020-06-22-R2:1800
```

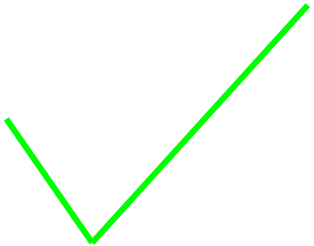
All files must be either tab-delimited in .asc, .tsv, .txt or…

variable	strain	name	media	condition	techrep	biorep
A1	none	blank	PR	(-Fe)	1	
A2	none	blank	PR	(10Fe)	1	
A3	none	blank	PR	(100Fe)	1	
A4	none	blank	PR	(250Fe)	1	
A5	H26	pyrE	PR	(--Fe)	1	1
A6	H26	pyrE	PR	(--Fe)	2	1
A7	H26	pyrE	PR	(--Fe)	3	1
A8	H26	pyrE	PR	(--Fe)	1	2
A9	H26	pyrE	PR	(--Fe)	2	2
A10	H26	pyrE	PR	(--Fe)	3	2
A11	H26	pyrE	PR	(--Fe)	1	3
A12	H26	pyrE	PR	(--Fe)	2	3
B1	H26	pyrE	PR	(--Fe)	3	3
B2	SH118	idr	PR	(--Fe)	1	1
B3	SH118	idr	PR	(--Fe)	2	1
B4	SH118	idr	PR	(--Fe)	3	1
B5	SH118	idr	PR	(--Fe)	1	2
B6	SH118	idr	PR	(--Fe)	2	2
B7	SH118	idr	PR	(--Fe)	3	2
B8	SH118	idr	PR	(--Fe)	1	3
B9	SH118	idr	PR	(--Fe)	2	3
B10	SH118	idr	PR	(--Fe)	3	3
B11	SH119	sirR	PR	(--Fe)	1	1
B12	SH119	sirR	PR	(--Fe)	2	1
C1	SH119	sirR	PR	(--Fe)	3	1
C2	SH119	sirR	PR	(--Fe)	1	2
C3	SH119	sirR	PR	(--Fe)	2	2
C4	SH119	sirR	PR	(--Fe)	3	2
C5	SH119	sirR	PR	(--Fe)	1	3



Files can also be in .csv

```
Label,Info,variable,strain,name,media,condition,techrep,biorep
,,Blank,,,,,,
,,1,none,blank,PR,(-Fe),1,
,,2,none,blank,PR,(10Fe),1,
,,3,none,blank,PR,(100Fe),1,
,,4,none,blank,PR,(250Fe),1,
,,5,H26,pyrE,PR,(--Fe),1,1
,,6,H26,pyrE,PR,(--Fe),2,1
,,7,H26,pyrE,PR,(--Fe),3,1
,,8,H26,pyrE,PR,(--Fe),1,2
,,9,H26,pyrE,PR,(--Fe),2,2
,,10,H26,pyrE,PR,(--Fe),3,2
,,11,H26,pyrE,PR,(--Fe),1,3
,,12,H26,pyrE,PR,(--Fe),2,3
,,13,H26,pyrE,PR,(--Fe),3,3
,,14,SH118,idr,PR,(--Fe),1,1
,,15,SH118,idr,PR,(--Fe),2,1
,,16,SH118,idr,PR,(--Fe),3,1
,,17,SH118,idr,PR,(--Fe),1,2
,,18,SH118,idr,PR,(--Fe),2,2
,,19,SH118,idr,PR,(--Fe),3,2
,,20,SH118,idr,PR,(--Fe),1,3
,,21,SH118,idr,PR,(--Fe),2,3
,,22,SH118,idr,PR,(--Fe),3,3
,,23,SH119,sirR,PR,(--Fe),1,1
,,24,SH119,sirR,PR,(--Fe),2,1
,,25,SH119,sirR,PR,(--Fe),3,1
,,26,SH119,sirR,PR,(--Fe),1,2
,,27,SH119,sirR,PR,(--Fe),2,2
,,28,SH119,sirR,PR,(--Fe),3,2
```



Ensure Data Files Match Corresponding Mapping Files

Data File:

Time (default: 600s)

Well Position

Time	0	600	1200	1800	2400	…	59400
A1	0.277	0.275	0.279	0.280	0.289	…	0.474
B1	0.282	0.275	0.280	0.281	0.289	…	0.478
C1	0.279	0.276	0.278	0.282	0.289	…	0.478
D1	0.267	0.266	0.271	0.274	0.282	…	0.416
…	…	…	…	…	…	…	…
E12	0.303	0.295	0.301	0.309	0.314	…	0.713
F12	0.285	0.299	0.313	0.310	0.305	…	0.667
G12	0.278	0.297	0.313	0.333	0.349	…	0.753
H12	0.265	0.270	0.271	0.291	0.293	…	0.614

Well positions must be “Letter + Number” format;
can go beyond H12 (if using more than 96 wells).

Mapping File:

Well	Plate_ID	Substrate	Ribotype
A1	2020-06-22-R1	Glucose	RT078
B1	2020-06-22-R1	Glucose	RT078
C1	2020-06-22-R1	Fructose	RT078
D1	2020-06-22-R1	Fructose	RT078
…	…	…	…
E12	2020-06-22-R1	Trehalose	RT078
F12	2020-06-22-R1	Trehalose	RT078
G12	2020-06-22-R1	Water	RT078
H12	2020-06-22-R1	Water	RT078

If Using Metafile or Time Interval File

Mapping File: “exmapping1.txt”

Well	Plate_ID	Substrate	Ribotype
A1	2020-06-22-R1	Glucose	RT078
B1	2020-06-22-R1	Glucose	RT078
C1	2020-06-22-R1	Fructose	RT078
D1	2020-06-22-R1	Fructose	RT078
…	…	…	…
E12	2020-06-22-R1	Trehalose	RT078
F12	2020-06-22-R1	Trehalose	RT078
G12	2020-06-22-R1	Water	RT078
H12	2020-06-22-R1	Water	RT078

Meta file: “exmeta.txt”

Plate_ID	Species
2020-06-22-R1	E_randomii

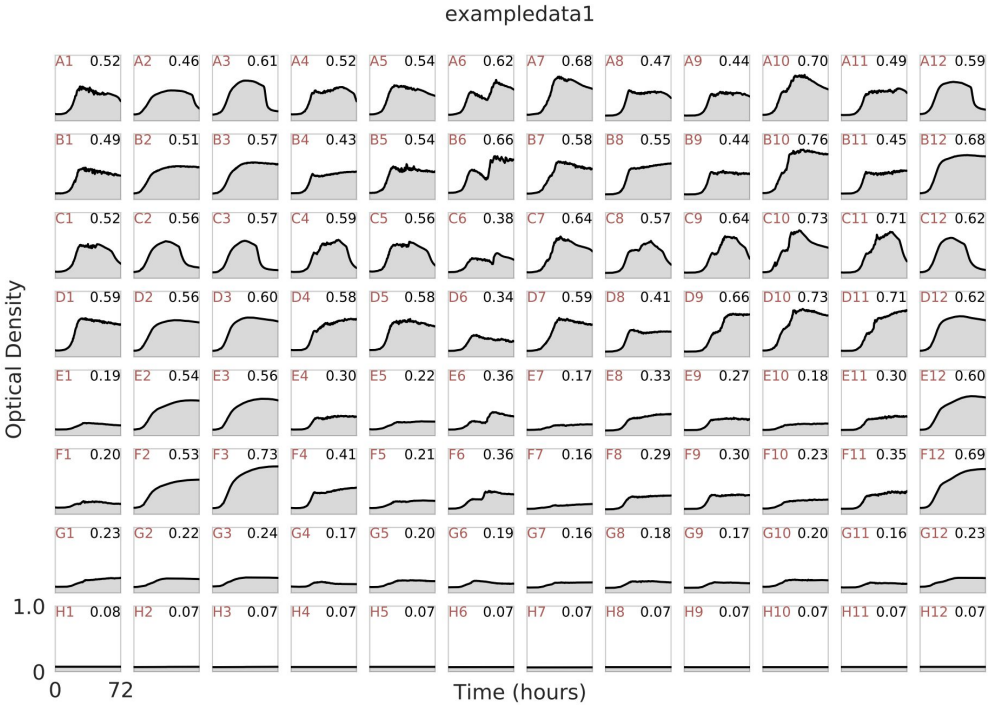
This will be a new column assigned to every well in datafile

Time interval file: “exinterval.txt”

```
Plate_ID:Interval(seconds)
2020-06-22-R1:1200
2020-06-22-R2:1800
```

Ensure that Plate_ID are matching between mapping, meta, and time interval.

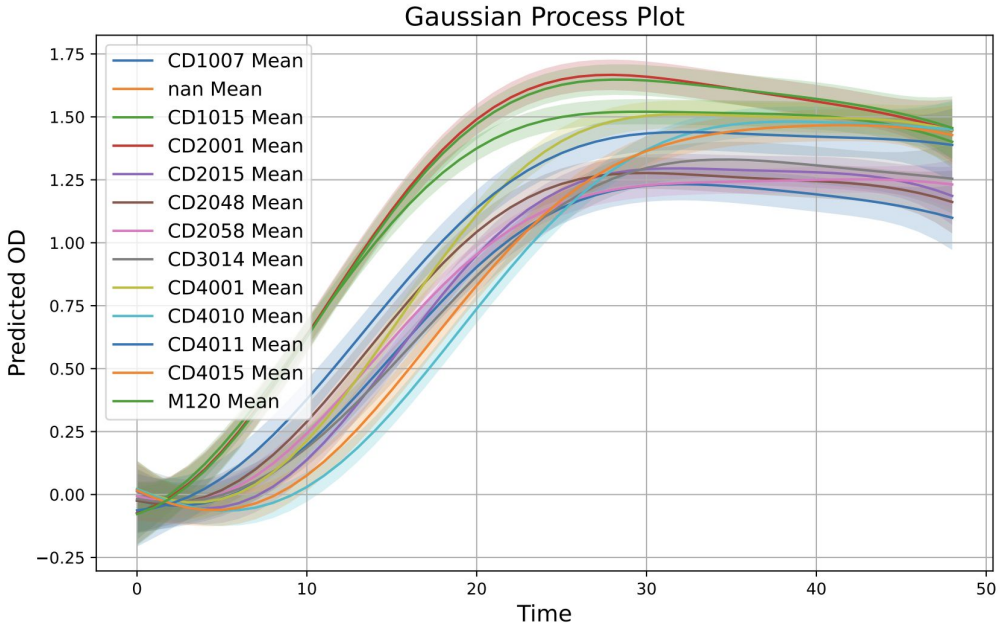
Raw Data Plotting



- Generate a **growth curve plot** for **each well** in the user's dataset.
  - Job would **only work with 96-well data** (A1~H12).
  - If you want to see generate raw data plots and have less than 96 wells, you can put in arbitrary OD readings (e.g. OD = 0.01) for empty wells.
- Number in the top right represents the fold change in optical density (OD) for each well relative to the control well.
- Fold change higher than 1.5 is in blue or and those lower than 0.5 is in red.

[Here for more detailed documentation on back end computing]

Gaussian Process Regression Curves Plotting



- This feature is more robust than raw data plotting and is not constrained by well numbers.
- For each option that user **pool** it will generate an growth curve based on each level of the pooling variable**.**
- User can also **subset** data.

Plot generated with example data though pooling by isolate (CD1007, etc)

Fit Statistics

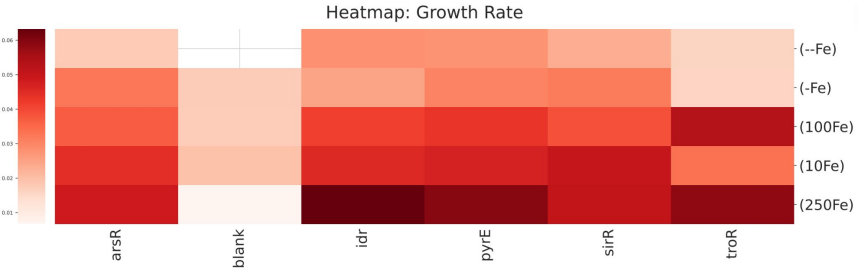
Substrate	Ribotype	AUC in units of OD	Linear Carrying Capacity in units of OD	Growth Rate	Doubling Time	Lag Time
Glucose	RT078	5.802649675463954	0.40810668	0.3948751654820088	1.7553577463243284	4.157499735213946
Fructose	RT078	5.6776837532211255	0.3811264713560038	0.3851002208559119	1.7999137445815474	4.309493492262543
Trehalose	RT078	5.989992162491322	0.4136694709584558	0.4169602566363442	1.6623818925852207	4.520302448160211
…	…	…	…	…	…	…
Trehalose	RT053	1.326368329466405	0.09668569	0.1531923277667863	4.524685998734635	5.174055881884366

example data: pool by Substrate + Ribotype will give me all the unique combinations of the two (values are arbitrary).

User can choose to pool data and five summary statistics of their GP curves will be generated:

1. area under the curve (AUC)
2. growth rate
3. linear carrying capacity
4. lag time
5. doubling time

Generating Heatmap



x being “name” (different strains) and y being “condition” (Fe conditions) and value being growth rate.

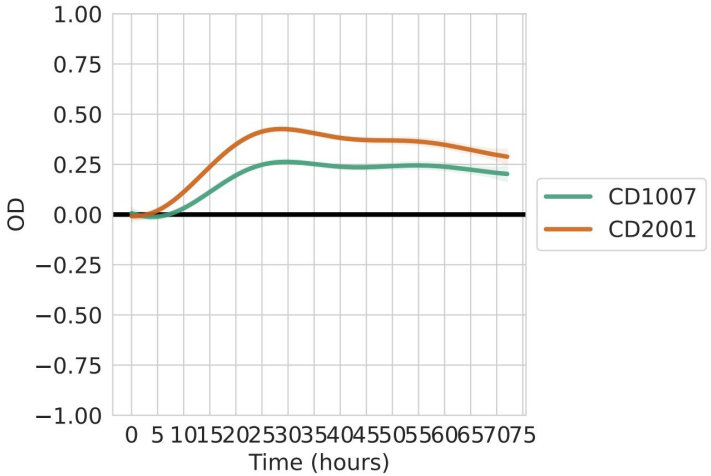
MicroNet allows users to generate heat maps. Users can choose columns in their data files as x-axis and y-axis. They can also choose one of the statistical values that they want to analyze: (AUC, Growth Rate, Carrying Capacity, lag time, doubling time).

Test Hypothesis

Example Summary:

The following criteria were used to subset data:
Isolate.....['CD1015', 'CD2001']
Model Tested: {'H0': ['Time'], 'H1': ['Time', 'Isolate']}
log Bayes Factor: 391.465
Functional Difference [95% CI]: 2.268 [1.710,2.825]

Example Plot:



Using GP regression to determine whether there is a significant difference in growth/OD between two specified conditions of an variable user select on.

Two outputs:

1. Test summary result: **Bayes Factor + functional differences**.
2. Plot of the normalized ln(OD) for the actual data + the predicted model for both conditions.

Here for more detailed documentation on back end computing

Compare Parameters

Substrate	Fructose	Trehalose	Fructose	Trehalose	
Parameter	**Mean**	**Mean**	**95.0% CI**	**95.0% CI**	**Sig. Diff.**
AUC (log)	26.655	19.261	[26.307,27.002]	[18.931,19.590]	TRUE
Death (log)	0.419	0.038	[0.331,0.508]	[-0.025,0.102]	TRUE
Diauxie	0	0	NA	NA	FALSE
Death Rate	-0.087	-0.019	[-0.140,-0.035]	[-0.072,0.035]	FALSE
Growth Rate	0.256	0.173	[0.237,0.274]	[0.156,0.191]	TRUE
Carrying Capacity (log)	1.586	1.146	[1.552,1.620]	[1.117,1.176]	TRUE
Lag Time	3.094	3.179	[2.862,3.325]	[2.837,3.522]	FALSE
Adaptation Time	0.242	0.348	[-0.595,1.078]	[-0.670,1.367]	FALSE
Time at Max. Death Rate	22.442	22.722	[18.889,25.994]	[19.480,25.964]	FALSE
Time at Max. Growth Rate	5.895	5.945	[5.481,6.309]	[5.430,6.460]	FALSE
Time at Carrying Capacity	14.303	20.312	[11.698,16.909]	[16.080,24.544]	FALSE
Doubling Time	2.712	4.011	[2.517,2.907]	[3.615,4.406]	TRUE

Directly compare the summary statistics between two specified conditions of an variable user select on (as opposed to determining differences in overall growth in Test Hypothesis)

Here for more detailed documentation on back end computing

Confirmation Email

You should get this immediately after you submitted a job.

MicroNet Job Confirmation

Hello,

Thank you for submitting a job with MicroNet. Your information is now being processed, and you will receive another email with your final results when the analysis has completed.

For more information, check out the links under the "Resources" page of MicroNet. Thank you for using MicroNet!

Regards, The MicroNet Team
schmidlabmodel@gmail.com

Results Email

If your job is successful, you should get all your files in an email.

MicroNet Analysis Results

Hello,

Attached is your final analyzed data.

For more information, check out the papers under the Resources page of MicroNet. Thank you for using MicroNet!

Regards, The MicroNet Team
schmidlabmodel@gmail.com

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heatmap_plot.pdf (application/pdf)	Download